

Maxwell's equations – physical interpretation

$$\oint \mathbf{E} \cdot d\mathbf{A} = Q/\epsilon_0$$

Gauss E: charges create E-field

$$\oint \mathbf{B} \cdot d\mathbf{A} = 0$$

Gauss B: no magnetic monopoles

$$\oint \mathbf{E} \cdot d\mathbf{l} = -d\Phi_B/dt$$

Faraday: changing B creates E

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 (\mathbf{J} + \epsilon_0 d\mathbf{E}/dt)$$

Ampere: current + changing E creates B

These four equations predict EM waves propagating at $c = 1/\sqrt{(\mu_0 \epsilon_0)} = 2.998 \times 10^8 \text{ m/s}$